

DVAS — ONE VOTER ID, ANYWHERE VOTING

Project Summary

Project Name	DVAS — One Voter ID, Anywhere Voting
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Problem Statement

Eligible voters who are away from their registered polling location may miss voting, while allowing access from another location without synchronized status can create a risk of duplicate voting.

Solution in One Line

DVAS verifies synthetic voter eligibility and central voting status, creates a short-lived one-time authorization, blocks repeat attempts across demo locations, and keeps voter identity separate from anonymous vote records.

Project Summary (232 words)

One Voter ID—Anywhere Voting improves electoral access while protecting security and privacy. Eligible voters may be away from their registered polling location on election day because of work, education, or travel. Our prototype demonstrates how an authorized location verifies synthetic voter identity, confirms eligibility, and checks central status before allowing one voting session.

When a voter is eligible and marked NOT_VOTED, the system issues a single-use session. Once voting is completed, status changes immediately to VOTED. If the same voter ID is attempted from another demo location, such as Surat after Ahmedabad, the system blocks the second attempt, preventing duplicate voting.

The project does not replace certified EVMs; the simulated voting screen exists only to demonstrate the end-to-end journey without connecting to live electoral infrastructure. Privacy is protected through decoupled data separation: the Voter Status Ledger stores synthetic identity and voting status, while the Anonymous Vote Vault records candidate selections with zero voter linkage.

Multilingual voice guidance in Gujarati, Hindi, and English ensures accessibility for first-time and elderly voters. Built-in AI tools provide staff guidance and real-time anomaly alerts without accessing ballot choices, delivering a secure, inclusive, and practical civic prototype.